PO12Q - Quantitative Political Analysis: Uncovering Relationships

Week 2 - Worksheet Solutions

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1. a. FALSE: A coefficient of −1 indicates a *perfect negative* correlation. It is a coefficient of 0 that indicates no correlation at all.

   b. TRUE: Pearson's correlation assumes a linear relationship, whereas Spearman's relies only on the relationship being monotonic — one variable consistently increasing or decreasing as the other changes, though not necessarily at a constant rate or in a straight line. Spearman's is therefore better suited to non-linear monotonic relationships.

   c. TRUE: The variance is the covariance of a variable with itself: setting $y = x$ in the covariance formula returns the formula for the variance.

   d. FALSE: The covariance is unbounded and depends on the measurement units of the variables. It is the *correlation coefficient* — the covariance normalised by the product of the standard deviations — that is bounded between −1 and +1.

   e. TRUE: Pearson's correlation is computed from the means and standard deviations of the raw data, both of which can be heavily affected by extreme values, making it sensitive to outliers.

   f. TRUE: Spearman's coefficient first ranks the values of each variable and then applies Pearson's correlation to those ranks.

   g. TRUE: The point-biserial correlation is a special case of Pearson's correlation, applied when one variable is continuous and the other is binary (coded 0/1).

   h. FALSE: The ϕ-coefficient measures the association between two *dichotomous (binary)* variables, not two continuous ones. Two continuous variables call for Pearson's (or Spearman's) correlation.

   i. TRUE: A coefficient of +1 indicates perfect positive correlation, meaning all observations lie exactly on a straight (upward-sloping) line.

   j. FALSE: The slope of the line does not matter. The correlation coefficient measures how tightly the observations cluster around a line, not how steep that line is.

   k. FALSE: A significant correlation establishes association only. Correlation is necessary but not sufficient for causation, so it can never demonstrate that one variable causes another on its own.

   l. TRUE: With a small sample, even a sizeable correlation may fail to reach significance because of low statistical power. A non-significant result then reflects insufficient sample size rather than the genuine absence of an association.

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2. a. Pearson's correlation — two continuous, approximately normally distributed variables with a presumed linear relationship.

   b. Point-biserial correlation — one continuous variable (fare) and one dichotomous variable (survived: yes/no).

   c. ϕ-correlation — two dichotomous variables (sex and survival status).

   d. Spearman's correlation — both variables are ordinal (ranks/finishing positions).

   e. Spearman's correlation — the outliers in income make Pearson's unreliable, whereas Spearman's, being rank-based, is robust to extreme values (and makes no normality assumption).

3. The table below records the number of hours five students spent revising (x) and their score on a test (y).

Student	1	2	3	4	5
x (hours)	2	4	5	7	8
y (score)	50	55	65	70	80

   a. $\bar{x} = 26/5 = 5.2$ and $\bar{y} = 320/5 = 64$.

   b. Intermediate calculations:

x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$	$(x - \bar{x})^2$	$(y - \bar{y})^2$
2	50	-3.2	-14	44.8	10.24	196
4	55	-1.2	-9	10.8	1.44	81
5	65	-0.2	1	-0.2	0.04	1
7	70	1.8	6	10.8	3.24	36
8	80	2.8	16	44.8	7.84	256
			Sum	111.0	22.80	570

$$r_p = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}} = \frac{111}{\sqrt{22.80 \times 570}} = \frac{111}{114} = 0.97$$

   c.

$$t = \frac{r_p\sqrt{n-2}}{\sqrt{1 - r_p^2}} = \frac{0.97\sqrt{3}}{\sqrt{1 - 0.94}} = 7.40$$

   With $n - 2 = 3$ degrees of freedom, the critical value for 95% confidence is 3.182. Since $7.40 > 3.182$, the coefficient is statistically significant.

   d. The coefficient is positive and very strong (≥ 0.7): students who spent more hours revising tended to achieve higher test scores.

4. Consider the following five paired observations.

Observation	1	2	3	4	5
x	5	10	15	20	25
y	12	18	15	30	28

a. Ranks and squared rank differences:

x	y	Rank(x)	Rank(y)	d_i	d_i^2
5	12	1	1	0	0
10	18	2	3	-1	1
15	15	3	2	1	1
20	30	4	5	-1	1
25	28	5	4	1	1
				Sum	4

b.

$$r_s = 1 - \frac{6\sum d_i^2}{n(n^2-1)} = 1 - \frac{6(4)}{5(25-1)} = 1 - \frac{24}{120} = 0.80$$

c. The coefficient is positive and very strong (≥ 0.7): higher values of x are associated with higher values of y in a monotonic fashion.

5. Eight students sat a test. Four of them had completed a revision course (coded 1) and four had not (coded 0). Their scores were:

- Completed the course (group 1): 8, 9, 7, 8
- Did not complete the course (group 0): 5, 4, 6, 5

The standard deviation of all eight scores is 1.77.

a. $\bar{x}_1 = (8 + 9 + 7 + 8)/4 = 8$ (completed the course); $\bar{x}_0 = (5 + 4 + 6 + 5)/4 = 5$ (did not).

b. With $n_1 = n_0 = 4$, $n = 8$, and $s = 1.77$:

$$r_{pb} = \frac{\bar{x}_1 - \bar{x}_0}{s}\sqrt{\frac{n_1 n_0}{n(n-1)}} = \frac{8-5}{1.77}\sqrt{\frac{4 \times 4}{8 \times 7}} = 1.69 \times 0.535 = 0.91$$

c. The coefficient is positive and very strong (≥ 0.7): completing the revision course is associated with higher test scores. (This is, of course, only an illustrative small sample.)

6. A study of 80 patients recorded whether each received a treatment (yes/no) and whether they recovered (yes/no). The results are shown below.

	Recovered (1)	Not recovered (0)	Total
Treated (1)	A = 30	B = 10	40
Untreated (0)	C = 15	D = 25	40
Total	45	35	80

a. With $A = 30$, $B = 10$, $C = 15$, and $D = 25$:

$$\phi = \frac{AD - BC}{\sqrt{(A + B)(C + D)(A + C)(B + D)}} = \frac{(30)(25) - (10)(15)}{\sqrt{(40)(40)(45)(35)}} = \frac{600}{\sqrt{2{,}520{,}000}} = 0.38$$

b.

$$\chi^2 = n\phi^2 = 80 \times 0.38^2 = 11.43$$

With 1 degree of freedom, the critical value for 95% confidence is 3.84. Since $11.43 > 3.84$, the association is statistically significant.

c. The coefficient is positive and of medium strength (≥ 0.3): receiving the treatment is associated with recovering.

7. Each of the following correlation coefficients was found to be statistically significant. For each, describe the direction and strength of the relationship between the two named variables, using the strength thresholds given in the text.

   a. $r_p = -0.62$: a *negative*, *strong* (≥ 0.5) relationship — more screen time is associated with fewer hours of sleep.

   b. $r_s = 0.08$: a *positive* but *negligible* relationship (< 0.1, classed as no association) — exam rank and seating-row rank are effectively unrelated.

   c. $r_{pb} = 0.35$: a *positive*, *medium* (≥ 0.3) relationship — season-ticket holders tend to attend more games.

   d. $\phi = -0.91$: a *negative*, *very strong* (≥ 0.7) relationship — vaccination status and infection status are strongly inversely associated (those vaccinated are much less likely to be infected).